MACHINE LEARNING MINI-PROJECT

Pest Prediction

Three models compared on 110,469 field-year records from Spain and California

[Your name] | [Guide name] | [Date] Data: Lippey et al. (2026), Dryad, doi:10.5061/dryad.ffbg79d87

Problem statement

Farmers usually notice a pest surge after the damage has started. Can a field's location, crop, pest species, local temperature and last year's count tell us in advance which field-years will have unusually high pest numbers?

Task: binary classification, High vs Normal abundance

110,469
pest field-year records, 30 pest populations

5 crops
cotton, olive, citrus, grape, rice in Spain and California

1997–2019
train on earlier years, test on 2017–2019

Data: Lippey et al. (2026), "Field data challenge predictions of universal crop pest proliferation under warming", Dryad, doi:10.5061/dryad.ffbg79d87

Project objectives

1. Define a fair “high abundance” label that works across 30 different pest species
2. Train and compare three models on the same time-based split
3. Test whether last year’s count adds power beyond location and temperature
4. Tune hyperparameters and check whether tuning helps on unseen years
5. Report honest, measured results against simple baselines

Success criterion: beat random guessing (PR-AUC 0.234) on the 2017–2019 test years.

Methodology and data processing

Load	141,562 rows × 14 columns. One row is one species in one field in one year: mean abundance plus five temperature summaries.
Clean	Removed 7 exact duplicates and the 13 natural-enemy populations (31,086 rows). 110,469 pest rows remain; 637 lack temperature and are median-filled.
Label	High means above that species' own 75th percentile in the training years, because abundance scales differ hugely between species. 23.4% of test rows are High.
Features	Species, crop, region, subregion and five temperature values. Set B adds last year's count for the same field and species (available for 41% of rows). Spray counts are not used.
Split	Train on 1997–2016 (86,819 rows), test on 2017–2019 (23,650 rows). Medians, encoders and thresholds are learned from training rows only.

System design



Leakage safeguards built into the design

- Labels and thresholds come from training years only
- Imputer and encoder are fitted inside the pipeline on training rows
- Test years stay untouched until the final check

Experimentation workflow

1. Fit each model with default settings
2. Tune: 8 random settings, 3-fold cross-validation by year, on a 40,000-row sample
3. Refit the best setting on all training rows
4. Score once on the 2017–2019 test years

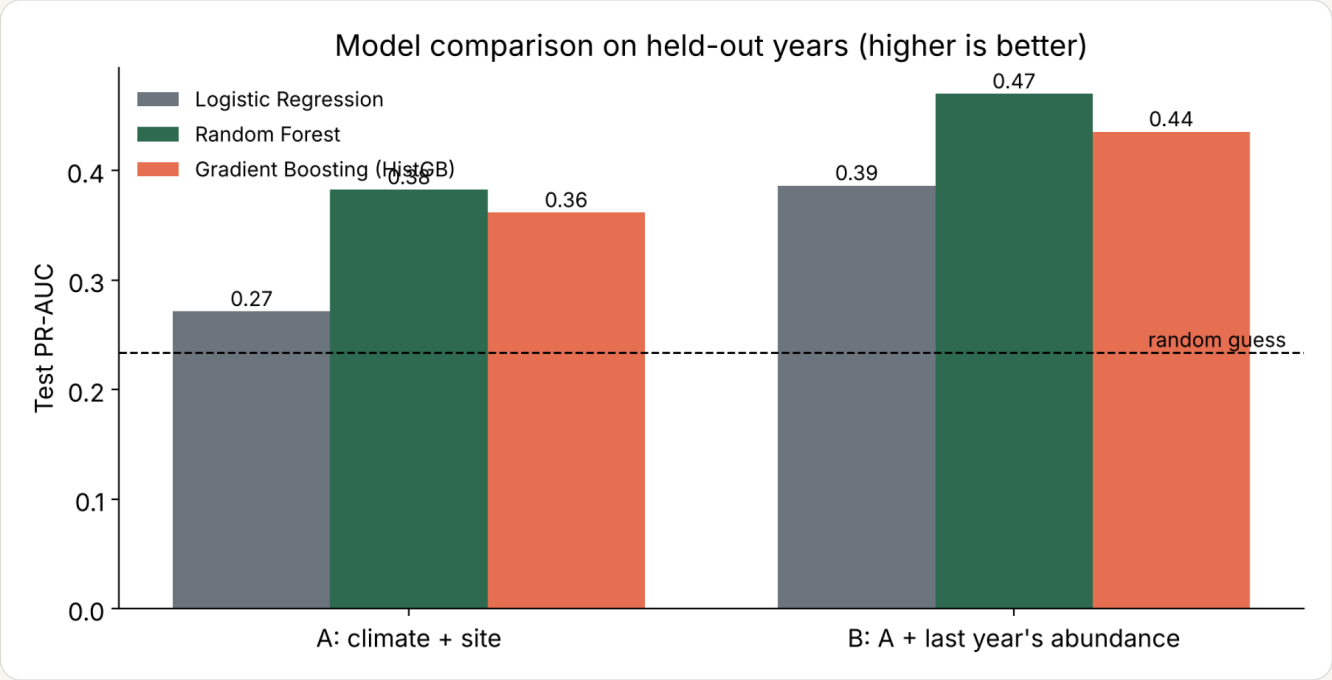
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experiments = 2 feature sets × 3 models

Set A: species, crop, location and five temperatures.
Set B: set A plus last year's count in the same field.

Selection rule: the exported model has the best cross-validated PR-AUC, fixed before the test set was opened.

Model comparison



Random Forest wins in both feature sets.

Last year's count lifts it from 0.383 to 0.470 PR-AUC.

Climate only with Logistic Regression (0.271) barely beats random guessing (0.234).

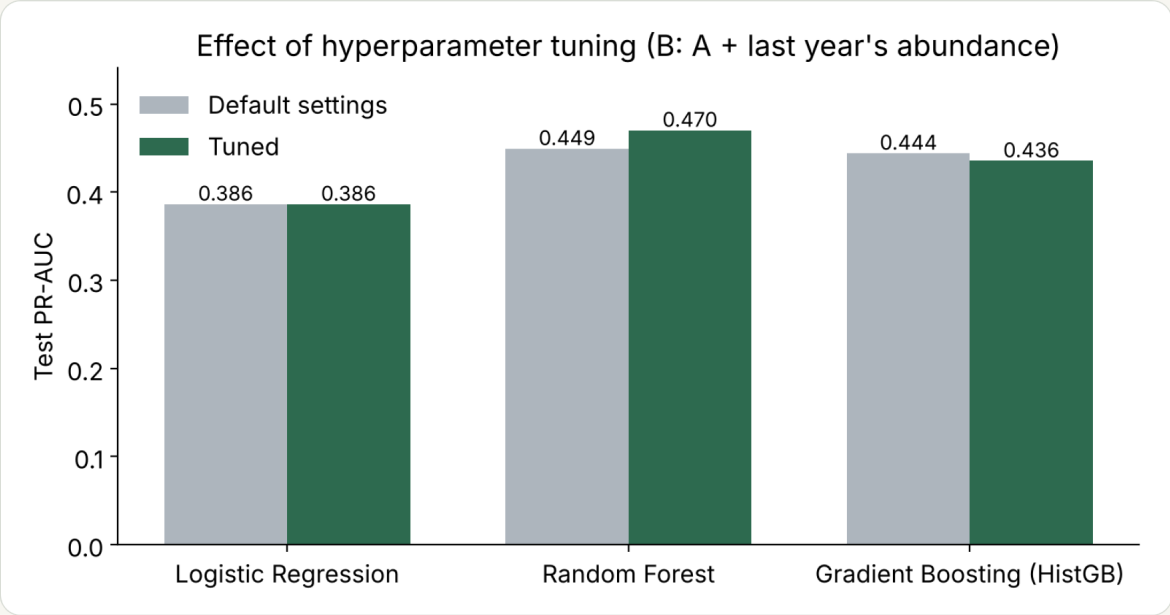
Test PR-AUC after tuning, on the 2017–2019 test years. Set A: climate and site. Set B: set A plus last year's count.

Results on unseen years

Set	Model	PR-AUC	ROC-AUC	Precision	Recall	F1
A	Logistic Regression	0.271	0.569	0.283	0.405	0.333
A	Random Forest	0.383	0.675	0.374	0.399	0.386
A	Gradient Boosting (HistGB)	0.362	0.659	0.323	0.580	0.415
B	Logistic Regression	0.386	0.674	0.363	0.423	0.391
B	Random Forest	0.470	0.748	0.439	0.440	0.440
B	Gradient Boosting (HistGB)	0.436	0.719	0.368	0.633	0.465

Set A = climate and site. Set B = set A plus last year’s count. Precision, recall and F1 use the default cut-off of 0.5. Random guessing scores PR-AUC 0.234; always predicting “Normal” is 76.6% accurate but finds no High years.

Hyperparameter tuning



Model	Searched	Best (set B)
Logistic Regression	C from 0.01 to 100	C = 0.01
Random Forest	Trees, depth, leaf size, features per split	300 trees, no depth limit, 30 per leaf, square-root features
Gradient Boosting	Rounds, depth, learning rate, L2, leaf size	100 rounds, depth 8, rate 0.03, L2 1.0, leaf 20

Tuning helped only once: Random Forest with last year's count (0.449 to 0.470). For the other models it changed little or slightly lowered the test score.

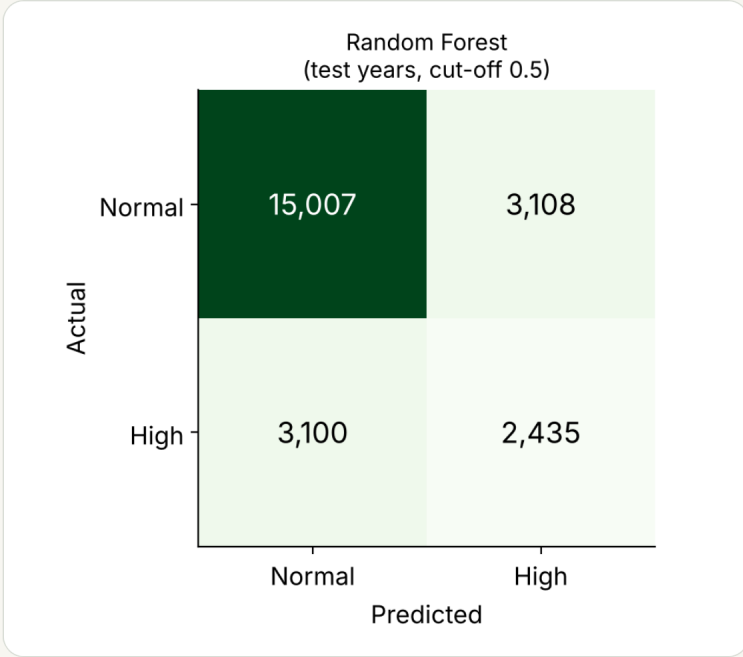
Final result

Random Forest with climate, site and last year’s count

Selected by the best cross-validated PR-AUC (0.437)

0.470	0.748	0.440
PR-AUC	ROC-AUC	F1 at cut-off 0.5

At a lower cut-off of 0.405 the model finds **83%** of High field-years, but **65%** of its alerts are false alarms.



Test set: 23,650 field-years, 5,535 truly High. Approximate 95% range for PR-AUC: 0.454 to 0.483.

Conclusion

Weather alone is a weak signal

With location and temperature only, the best model reaches PR-AUC 0.383 against 0.234 for random guessing.

Last year’s count helps most

It lifts the Random Forest to PR-AUC 0.470 and ROC-AUC 0.748 on unseen years.

A simple rule is hard to beat

Where last year’s count exists, “same as last year” scores F1 0.58, the same as the model.

Use with care

One benchmark dataset, labels relative to each species, no field trial. It should support scouting, not replace an agronomist.

Future work: add weekly weather, spray records and one model per species.